

Intro

As autonomous vehicles gain relevance and popularity, building a well rounded and reliable machine learning framework to balance the multiple dynamic objectives of driving is extremely important. Robustness, the ability to respond to varying situations, is a key factor in the success and safety of autonomous vehicles; as an algorithm needs to be able to successfully and efficiently navigate a wide variety of road configurations. This project investigates whether various pre-established evolutionary algorithms can be used to build a more robust model for autonomous driving. If an established algorithm such as Non-Dominated Sorting Genetic Algorithm (NSGA) (Kuo et al., 2023) is combined with a dynamic component (Ahrari et al., 2021), then a robust autonomous driving framework will be developed as it will be able to respond to a variety of situations in the most optimal way. We will simulate this framework with a small line following vehicle, representing an autonomous vehicle, which feeds data into our algorithm to be optimized across varying tracks. The NSGA element works by simulating a gene pool evolving and mutating to an environment in order to achieve Pareto optimization, where no optimization can be made to an objective without harming another (Liu et al., 2025). Including a dynamic element to form a dynamic multiobjective problem increases robustness (Ahrari et al., 2021) by exposing the model to more situations such as steeper or narrower curves, and training the NSGA algorithm in a way that measures the dynamic element as an extra dimension. The development of this framework will show a viable solution to the creation of new, more adaptable, autonomous driving models using new techniques.

Materials

- Hardboard pegboard (4ft by 8ft)
- Micromouse - Zumo Robot for Arduino (Pololu)
- Arduino UNO Rev3
- Plotter printer paper 6.25ft by 3.5 ft
- Plotter printer
- Laptop for code
- 4 AA batteries
- Arduino cable
- Extension cord

Methods

- 1) Selected independent variables to optimize for the traversal (movement efficiency and speed) of our micromouse: kp (proportional gain; the system's immediate reaction to current error), kd (derivative gain; predicts future error by analyzing rate of change), base speed (reg. speed of robot), min speed (lowest speed it is allowed to reach), stutter about each curve (the allotted amount the micromouse is allowed to "stutter" for to get back to original position), and brake power (how forceful the brakes are).
- 2) Code was created for the micromouse that allows for adjustable kp, kd, base speed, min. base speed, corner 1 2 & 3 stutter (for each turn on the track), and brake power. We created a default setting for the first generation of track 1 which would be modified by an NSGA (non-dominated sorting genetic algorithm) to produce optimal results over generations.
- 3) Varying tracks were designed, each with the same structure but with different sharpness of curves and printed them using a plotter printer, then we mounted them to the pegboard for testing.
- 4) The micromouse was plugged into the computer that has access to the code to allow the micromouse to run.
- 5) Before each run we calibrated the micromouse to the black line of the track. Calibration allows for the reflectance sensor array to track the line properly.
- 6) The micromouse was ran on track 1 to gain data for 10 generations of the NSGA algorithm which attempted to optimize the independent variables each generation for improved run time and veer score.
- 7) We took this 10 dimensional dataset and visualized it using a dimensionality reduction method, UMAP (Uniform Manifold Approximation and Projection for Dimension Reduction) to make figures that are visually digestible in 2 or 3 dimensions.
- 8) We repeated step 6 for tracks 2 and 3 to generate new data to using the NSGA algorithm, this created more datasets to be visualized and analyzed.
- 9) Finally we compared and contrasted our results to construct a conclusion on the efficacy of our optimization pipeline and how the NSGA algorithm responded to the slightly different tracks. This helped us build a mathematical model of Pareto optimal points and conduct data analysis.

Using Dynamic Non-Dominated Sorting Genetic Algorithm to Optimize Vehicle Locomotion Along Varying Curves

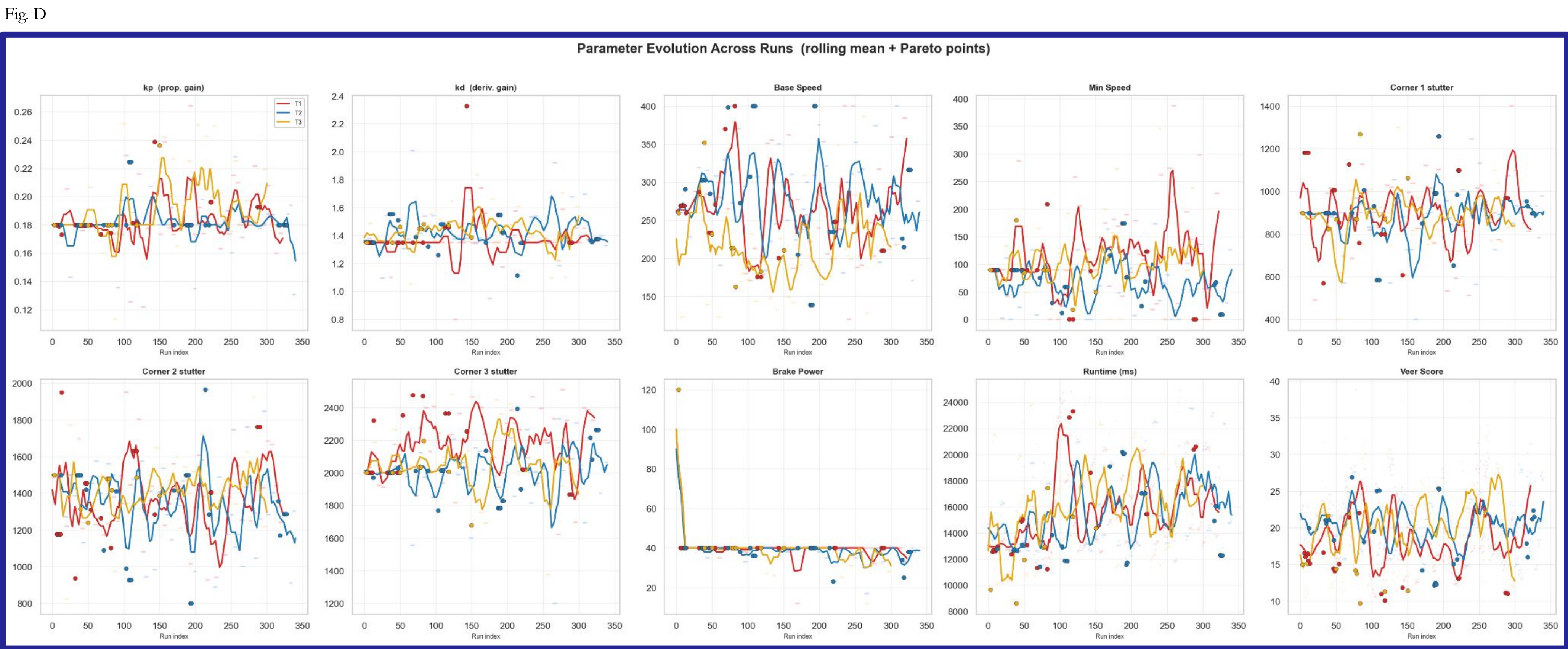
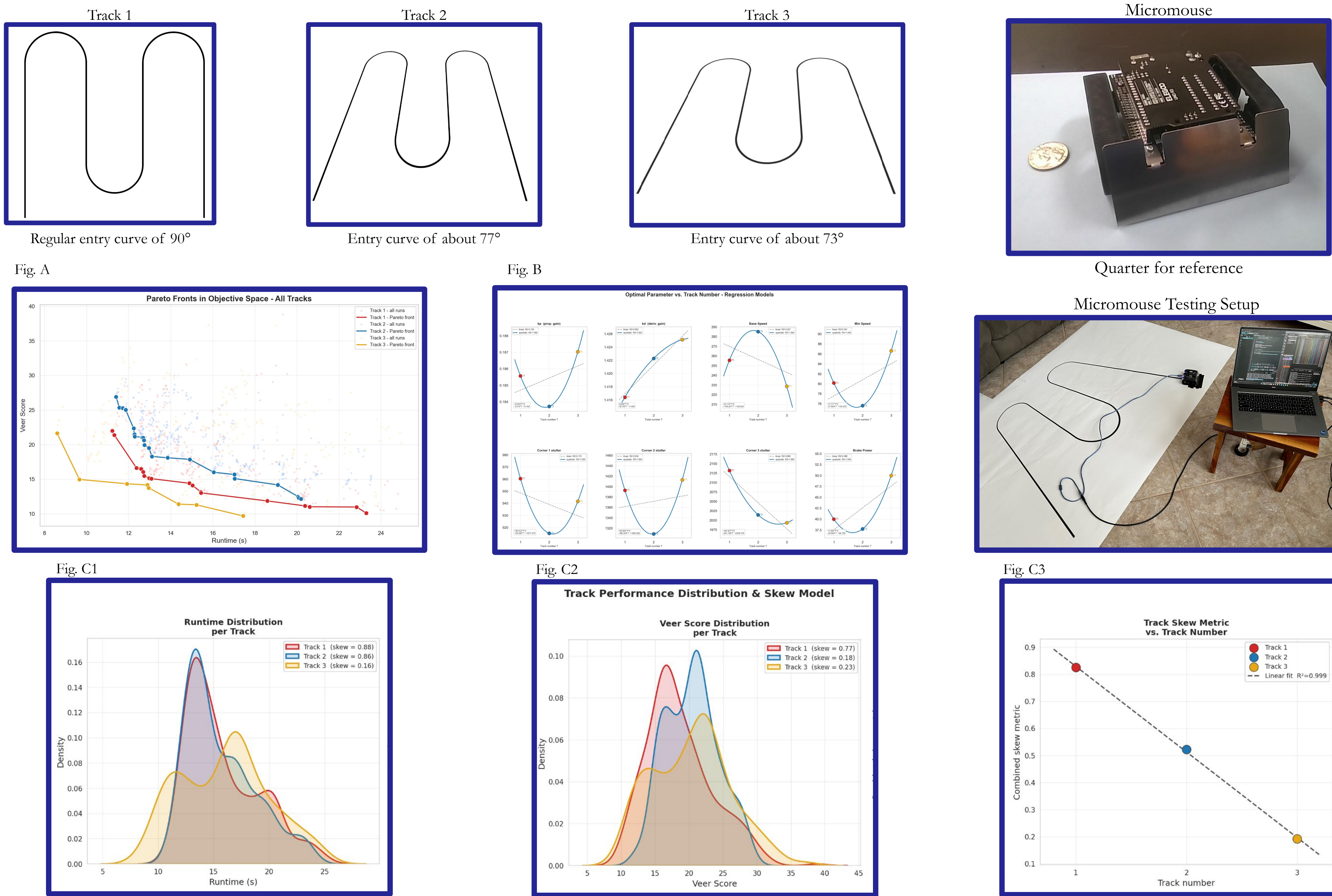
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Research Question

Can combining a pre-established evolutionary algorithm (Non-Dominated Sorting Genetic Algorithm) with a dynamic multiobjective component produce a more adaptable and robust machine learning framework for autonomous driving across different road configurations.

Hypothesis

If NSGA (Non-Dominated Sorting Genetic Algorithm) is integrated with a dynamic multiobjective component, then the resulting autonomous driving framework will be more robust and adaptable, as it will be able to respond optimally to a wider range of dynamic driving conditions such as steeper or narrower curves through Pareto-based optimization. In our implementation with a micromouse, faster run times and lower veer scores (average distance from the center line) across tracks of different corner tightness are the two goals NSGA attempts to balance.



Conclusion & Discussion

Using NSGA-II across three tracks of increasing curve sharpness, the framework we developed was able to optimize a multi-objective, multi-parameter control system and demonstrated the relationship between track geometry and the solution set. Track 3's sharper turns produced only 8 Pareto optimal data points compared to Track 1 and 2's 18 and 24 however they were higher quality with lower average run time and veer score. The downward trend in the combined Fisher skewness and the parameter evolution graphs show that sharper turns reduced performance variation and pushed the algorithm's optimization capability more by not giving it an easy solution to converge on which occurred for Track 1 and 2. Additionally the regression models built between Pareto optimal values across track numbers provide a mathematical framework for predicting ideal settings at untested track difficulty levels. This validates the NSGA-II pipeline as an effective tuning method for autonomous systems across dynamic scenarios, proving our hypothesis that it would be applicable, effective, and robust as it was able to find optimal solutions across varying conditions. Potential errors may have occurred with mechanical wear, especially wear on the motors forcing us to replace the motors multiple times, tread wear may have also decreased the traction of the micromouse over time, possibly creating higher veer scores. Additionally, we only collected data across three tracks which means our regression analysis is relatively weak and should be tested further.

Conclusion Cont.

Figure A, Pareto Fronts in Objective Space: A plot of finished runs on a graph with the x-axis being completion time and y-axis being veer score. Shows the tradeoff between the two objectives. Moreover we identified the points on the Pareto front, where runs cannot improve either veer score or completion time without compromising the other. The points that are Pareto optimal for each track are connected to form the front.

Figure B, Regression Models: Builds regression models between the different Pareto optimal values for each parameter across the tracks, as there is only three data points in the regression model there was always a quadratic fit that could connect all of the points.

Figure C, Skew Model: Measurement of how asymmetrical the distribution of run times and veer scores is for each track using Fisher skewness (where 0 is a perfectly symmetrical histogram). The linear graph in Figure C3, compares the combined skew metric which is the average of the Fisher skewness of both the run time histogram (C1) and the veer score histogram (C2) of different tracks.

Figure D, Parameter Evolution - Graphs the evolution of the values connecting rolling means of the values each parameter was set to across runs, as well as visualizing the Pareto optimal values. The consistent spread of Pareto optimal values across the runs confirms the improvement of the algorithm over time as well as visualizing the "thought process" of the NSGA algorithm when testing different generations of a parameter.

Future Work

Future research can expand on this project by testing the algorithm on a wider variety of track designs, including sharper turns, longer courses, intersections, unpredictable obstacles, to better simulate real world conditions. Increasing the number of generations and trials would also provide more data to create a more robust model. Additional control variables such as acceleration limits, sensor sensitivity, wheel traction and energy efficiency could also be included in the optimization process as well as using a model more adapted to multiple objectives such as NSGA-III which maintains diversity much better compared to NSGA-II to make the model more complex and realistic. This framework may be extremely useful for complex control system tuning, where multiple objectives must be balanced at the same time. Especially as systems become more and more complex and interconnected, traditional manual trial and error tuning becomes increasingly time consuming and less effective. The dynamic NSGA approach can automatically search and test many parameter combinations and produce Pareto optimal solutions that perform well across competing objectives. Comparing this framework with other machine learning methods such as reinforcement learning or neural networks, would further help evaluate its strengths and weaknesses. Future studies could also implement an instantaneous feedback system where a machine can continuously retune itself using a more efficient NSGA algorithm that continuously modifies the Pareto optimal solutions, making this framework more practical. Finally implementing this model on larger, faster, and more reliable robotic platforms could improve the practical relevance of this framework and help demonstrate its potential uses in full-scale autonomous vehicles or other robotic systems.

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